Fpt University

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CAPSTONE PROJECT REPORT

Report 5 – Software Test Documentation

– Ho Chi Minh, October 2026 –

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I. Record of Changes

Date	A* M, D	In charge	Change Description
20 October 2026	M	Project team	Prepared Review 2 Test Plan and template-based unit and integration/system test workbooks.

*A - Added M - Modified D - Deleted

II. Testing Documentation

1. Scope of Testing

Testing covers authentication and role scope, user media, mission planning and review, waypoint and segment validation, telemetry ingestion and monitoring, incident handling, post-flight reports and advisory AI. Low-level flight-control algorithms and aviation certification are outside the software test scope.

Unit tests exercise business rules and processing functions with deterministic inputs. Integration tests cover API, persistence, scope and provider boundaries. System tests cover the two-role mainflows and recovery scenarios. Flow owners prepare tests; another team member reviews the cases and evidence.

Tests use a disposable database, scoped test accounts, configured route and restriction fixtures, timestamped telemetry replay, and provider/email stubs. Live MAVLink, provider and email behavior require separate integration evidence. The case designs are prepared; no execution results are recorded in this document.

2. Test Strategy

The strategy combines functional and boundary tests, authorization checks, transaction and recovery tests, realtime measurements and usability checks. Each test is linked to its function, controlled inputs, expected result and execution evidence.

2.1 Testing Types

Functional tests use normal, invalid and boundary inputs to verify required outcomes. Authorization tests use distinct Operators and a Manager to verify role and record scope. Reliability tests use duplicate/out-of-order samples, database failure, missing packets and provider errors. Performance tests record latency, loss and workload conditions. Usability tests cover visible feedback, keyboard navigation and desktop/narrow-screen layouts. A case passes only when its expected outcome and evidence agree.

2.2 Test Levels

Unit: deterministic rule, transaction, aggregation and response-processing checks owned by the function author and peer reviewer. Integration: disposable-database/API and provider-stub scenarios owned by the flow owner. System: a tagged Web/API build and controlled telemetry source exercised by the team. Acceptance: nominated mainflows reviewed against agreed requirements with supervisors. The matrix below shows the applicable testing levels.

Type of Tests	Test Level			
	Unit	Integration	System	Acceptance
Functional and boundary	X	X	X	X
Authorization and data protection	X	X	X	X
Reliability and recovery	X	X	X	X
Performance and realtime latency		X	X	
Usability and compatibility			X	X

2.3 Supporting Tools

The tools below support repeatable checks and evidence collection. Versions identify the project configuration; actual run versions, build commit and environment are recorded with each execution.

Purpose	Tool	Vendor/In-house	Version
Backend build and static checks	dotnet build / dotnet format	Microsoft	.NET 8; SDK 8.0.400 with configured roll-forward
Offline authorization checks	DroneOps.AuthChecks	In-house	Version recorded by Git commit
Frontend checks	ESLint / TypeScript / Node test runner	Open source / Node.js	Project lockfile; Node 24 in CI
API integration and data fixtures	API client / disposable PostgreSQL / replay source	In-house / PostgreSQL	Recorded in each run
Test tracking and results	Template XLS / XLSX workbooks	Course templates	Review 2 submission

3. Test Plan

3.1 Human Resources

Each flow owner prepares fixtures, expected outcomes and execution evidence for the assigned mainflow. Peer review is performed by a different member. The team coordinates regression testing and review readiness.

Worker/Doer	Role	Specific Responsibilities/Comments
Ha Quang Nghia	MF1 owner	Mission review, validation, transaction and scope fixtures and tests.
Le Quoc Thong	MF2 owner	Telemetry, incident, timer, reconnect and readiness tests; FE/CI and dossier coordination.
Dang Dao Anh Duc	MF3 owner	History, report, playback and analytics expected-data fixtures.
Giang Nguyen Gia Khanh	MF4 owner	AI context, provider, provenance and failure tests.
Peer reviewer / supervisors	Review and acceptance	Review requirements, cases, evidence and unresolved defects.

3.2 Test Environment

Testing uses a tagged build, disposable PostgreSQL data, separate role-scoped accounts, deterministic route/zone/telemetry fixtures and controlled AI/email responses. Server credentials are supplied through the test environment. Actual machine, browser and migration versions are recorded with the test run.

Purpose	Tool	Provider	Version

Backend service	.NET API + configured test environment	Microsoft / project	.NET 8; run build commit recorded
Web client	Next.js / React / Node.js	Vercel / Meta / Node.js	Project lockfile; CI Node 24
Database	Disposable PostgreSQL schema	PostgreSQL / project	Migration level recorded per run
Browser	Chrome / Edge	Google / Microsoft	Exact version recorded per run
Telemetry and external services	Simulator/replay + AI/email stubs	Project / configured providers	Fixture, rule settings and provider scenario recorded

3.3 Test Milestones

Milestone dates are aligned with the confirmed project and Review 2 schedule. Integration and system execution begin when the relevant APIs, migrations and test fixtures are available.

Milestone Task	Start Date	End Date
Test design and peer review	20 October 2026	Before Review 2 submission
Unit checks and offline regression	After function/test implementation	Before integration gate
API/database integration tests	After API/migration fixture readiness	Before system test gate
System workflow and demo rehearsal	After integration gate	Before confirmed Review 2 session
Feedback regression	After review feedback is recorded	Before corresponding delivery gate

4. Test Cases

Unit cases follow the supplied function-oriented XLS workbook. Integration, system and acceptance cases follow the supplied Test Report workbook. Each case records its requirement, precondition/input, procedure or condition matrix, expected result, type and execution evidence.

- *Unit Test Cases: Report5_Unit Test.xlsx*

- *Other Test Cases (IT, ST, AT): Report5_Test Report.xlsx*

5. Test Reports

Executed regression subset: TC-T01, TC-T02, TC-T03, TC-H01 and TC-H04 passed in isolated automated service/API fixtures. UT-05 and UT-06 conditions also passed through the service-regression harness. Supporting runs: 38 auth/validation, 39 telemetry, 42 post-flight and 18 FE contract/render checks passed. Evidence includes commands, commit 9d2078867285a499d3a8d6cb9c112521ae9877a0 and raw logs in test-evidence. Results apply to this subset and environment, not to browser/hardware or full acceptance testing.

Automatic flight preparation and workflow ownership

Current ownership: MF1 / WF-01 — NghiaHQ (Ha Quang Nghia), planning and preflight; MF2 / WF-02 — ThongLQ (Le Quoc Thong), realtime monitoring, incidents and flight preparation; MF3 / WF-03 — DucDDA (Dang Dao Anh Duc), post-flight; MF4 / WF-04 — KhanhGNG (Giang Nguyen Gia Khanh), AI assessment. ThongLQ remains team leader and dossier coordinator.

Implemented scope

AutomaticFlightRequested is optional and defaults to false. Changes follow the mission proposal and approval workflow. This release prepares readiness; dispatchEnabled is false. Neither confirmation nor gateway enrollment uploads a mission, arms motors, sends TAKEOFF/AUTO/MISSION_START, or changes the mission to InProgress.

Confirmation conditions

For an approved, opted-in mission with a schedule, the backend opens confirmation at T minus 5 minutes and closes it at the scheduled start. The authenticated operator must keep a visible browser tab online. Presence is sent every 10 seconds and expires after 30 seconds. The selected, unrevoked mission connection must match the active UAV and flight-controller UID, with gateway heartbeat within 30 seconds and telemetry within 15 seconds.

Readiness evidence

Confirmation reruns preflight and binds readiness to the exact plan hash and connection ID. Readiness requires a passing preflight for that hash from the last minute. Changed plan/connection, stale presence or missing telemetry invalidates the conditions; the operator must review and confirm again. Dismissal, a hidden/offline tab or a missed window is not flight consent.

Future execution scope

Autopilot execution remains a separate development phase: firmware/airframe capability detection, mission compilation and upload/read-back, command IDs/deadlines and replay protection, scheduled condition rechecks, local pre-arm/failsafe enforcement, and ACK plus observed flight-state verification. Validate ArduPilot/PX4 SITL, propeller-free bench operation and supervised flight before enabling dispatch. AI advice cannot authorize launch.

Test execution status

The following flight-preparation cases are planned acceptance/regression targets. No execution date, pass result or physical-aircraft evidence is claimed by this document update. MF2 owner ThongLQ coordinates execution; NghiaHQ reviews preflight fixtures; KhanhGNG verifies that AI advice cannot bypass readiness.

AF-01 Opt-in review

Create a mission without the option, then propose an option change. Expect false by default and the approved plan unchanged until review.

AF-02 Window boundaries

Check immediately before T-5, at T-5, immediately before start, and at start using backend time. Expect confirmation only within [T-5, start).

AF-03 Identity and scope

Try another owner, wrong UAV/UID, revoked/expired connection and inactive UAV. Expect denial or blocked readiness; no aircraft command.

AF-04 Freshness

Exercise stale operator presence (>30 s), gateway heartbeat (>30 s) and telemetry (>15 s), and restore them independently. Missing prerequisites must remain visible; restoring one must not bypass the others.

AF-05 Evidence changes

Confirm, then change plan hash or connection, or expire the passing preflight (>1 minute). Expect earlier evidence not to authorize readiness for changed/stale inputs.

AF-06 Dismiss and disconnect

Dismiss/cancel; hide the tab; lose Internet/session; reconnect after the window. Expect no implied consent or late launch, and cancel clears saved confirmation/presence.

AF-07 Plan contract

Export as owner and non-owner; verify units, waypoint order, altitude reference, E7 conversion and dispatchEnabled=false. Reject unauthorized export; do not interpret a missing altitude as zero.

AF-08 No dispatch

Confirm with all prerequisites and inspect gateway traffic and mission state. Expect no mission upload, ARM, TAKEOFF, mode/start command or readiness-driven InProgress transition.